Fraction Interpretation — Achievers / HOTS — Answer Key

1. **A) 27** — 2/5 of 45 = 18 roses, so lilies = 45 − 18 = 27.
2. **A) 15** — 1/3 of 36 = 12 apples. 1/4 of 36 = 9 bananas. Oranges = 36 − 12 − 9 = 15.
3. **A) 2/3** — Using a common denominator of 15: 2/3 = 10/15 and 3/5 = 9/15. Since 10/15 > 9/15, 2/3 is greater.
4. **A) 3/4** — 1/2 = 2/4, so 1/2 + 1/4 = 2/4 + 1/4 = 3/4.
5. **A) 1/2** — 1/3 = 2/6, so 1/3 + 1/6 = 2/6 + 1/6 = 3/6, which simplifies to 1/2.
6. **A) 10** — 3/4 of 40 = 30 red items. 1/3 of 30 = 10 red circles.
7. **A) 5/8** — There are 16 shapes in total and 10 are circles: 10/16. The greatest common factor of 10 and 16 is 2, so 10÷2=5 and 16÷2=8, giving 5/8.
8. **A) 1/4, 1/2, 2/3** — Using twelfths: 1/2=6/12, 2/3=8/12, 1/4=3/12. From smallest to largest: 3/12, 6/12, 8/12 — that is 1/4, 1/2, 2/3.
9. **D** — A) 1 of 4 shaded = 1/4, not 1/2. B) 4 of 10 shaded = 4/10 = 2/5, not 1/2. C) 3 of 8 shaded = 3/8, not 1/2. D) 3 of 6 equal parts shaded = 3/6 = 1/2 — correct, even though it isn't split into just 2 parts. Only the diagram whose shaded fraction simplifies to exactly 1/2 is correct — it doesn't have to be split into just 2 parts.
10. **A) 9** — 5/8 of 24 = 15 m used, so 24 − 15 = 9 m are left.
11. **A) 4** — Add the wholes: 2+1=3. Add the fractions: 1/2+1/2=1 whole. So the total is 3+1 = 4.
12. **A) 2 3/4** — 11 ÷ 4 = 2 remainder 3, so 11/4 = 2 3/4 cups.
13. **A) 19** — 2/5 of 40 = 16 blue. 1/8 of 40 = 5 red. Green = 40 − 16 − 5 = 19.
14. **A) 7/10** — Circles = 5 + 2 = 7. Total shapes = 8 + 2 = 10. The new fraction is 7/10 (already in simplest form, since 7 and 10 share no common factor other than 1).
15. **D** — A) only 4 of 6 sectors shaded — that's 4/6 = 2/3, not 5/6. B) 4 of 5 sectors shaded — that's 4/5, a different total, not 5/6. C) 5 of the 6 parts are shaded by COUNT, but the parts are NOT equal size, so the shaded area is far less than 5/6. D) 5 of 6 EQUAL sectors shaded — that really is 5/6. Only the diagram with 5 of 6 truly EQUAL parts shaded correctly shows 5/6.

How to use this: 10+/15 is strong at this difficulty. For the multi-step word problems, work out and label EACH category's count before answering — don't try to do it all in your head. For the diagram questions, remember a fraction only matches the picture when the parts are truly equal in size.